

Optimizing Portering Logistics at HHH

AI-Driven Porter Dispatch System

Team

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Presentation Outline

1. The Problem

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2. Data Foundation

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3. Our Solution

CHOI

- System Architecture
- LLM Natural Language Interface
- OR-Tools VRPTW Optimizer
- RAG Policy Advisor
- End-to-End Demo

4. NLP Parsing Accuracy

KIM

5. Optimization Results

KIM

6. Conclusion & Future Work

KIM

The Problem: Porter Dispatch at HHH

41.3%

of tasks miss the internal pickup KPI

81,349 real HHH porter tasks (Jan – Dec 2024) · Mean duration: 46.3 min · KPI target: 15 min

MISS 41.3%

PASS
58.7%

PORTER ROLE

What is a Porter?

Hospital transport specialists moving patients, specimens, and supplies between ward, clinic, and lab locations.

81,349

porter tasks in 2024

89

hospital locations

8

service types handled

CURRENT METHOD

App-Based FIFO Dispatch

- 1 Task submitted into the system
- 2 System auto-assigns an available porter
- 3 Porter receives task via app
- 4 NFC tap on completion triggers next assignment cycle

*Assignment quality depends on:
who happens to be idle — not who is best positioned.*

STRUCTURAL GAPS

Why Dispatch Fails

No Fleet Visibility

Porter locations and real-time status is unavailable to the dispatcher

No Optimization

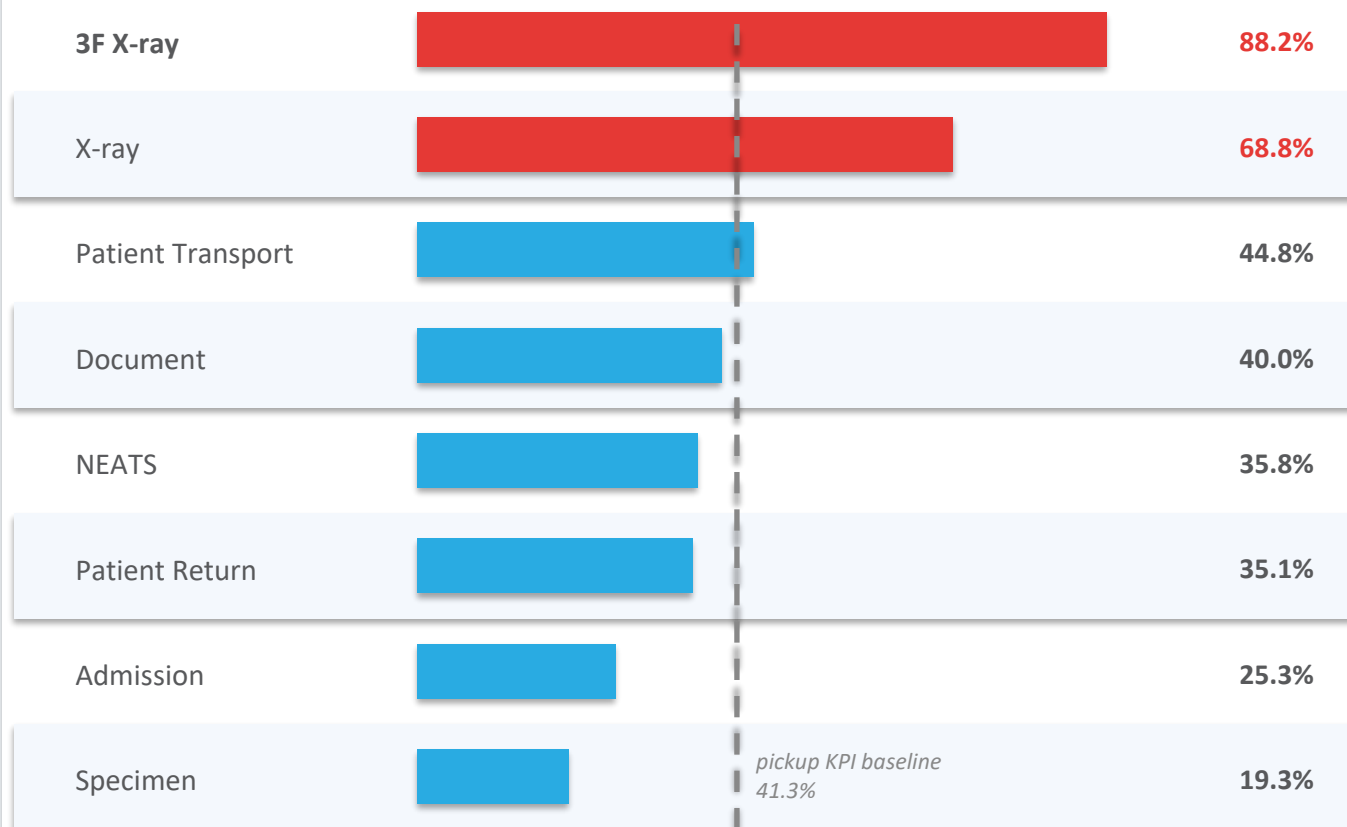
Only the nearest idle porter is selected; no KPI optimization is applied

No Factor Weighting

Travel distance to target, task duration, concurrent workload is not considered

Root Causes of Poor KPI Performance

PICKUP RESPONSE TIME VIOLATION RATE BY SERVICE TYPE



3F X-ray tops the chart (88.0%) because imaging suites are concentrated on specific floors — few porters are ever geographically close to the pickup origin when a task arrives.

ROOT CAUSES

Why KPI Performance Is Poor

FIFO approach — No Fleet-Wide Optimisation

- Order only — no location or fleet awareness
- A porter is dispatched locally without considering all pending tasks — missing better multi-porter solutions

Geographic Demand Concentration

- Imaging suites are concentrated on specific floors, leaving few porters near pickup origins
- 3F X-ray hits 88.2% pickup violation — highest of all service

Insufficient Fleet Size for KPI Compliance

- Task arrival rate exceeds porter completion rate at peak hours
- Dispatch optimisation alone cannot overcome a structural under-capacity in the fleet.

Data Foundation — 84,000 Real HHH Tasks

TASK DATASET — DATA2024.XLSX

Raw Data: ~84,000 rows · 50 columns · Jan – Dec 2024

Column	Field	Role in System
下單	Order Time	Task start · KPI response-time clock begins
從	Origin	Porter pickup location (1 of 89 nodes)
往	Destination	Delivery location
服務	Service Type	X-ray, Specimen, Document, Patient ...
優先	Priority	Urgent / Normal / Scheduled
感染控制	Infection Flag	Triggers isolated porter assignment
完成	Completion Time	Task end · used for total duration calc

TRAVEL TIME MATRIX

89 Hospital Locations · 7,921 Pairs

WHY p75?

Mean travel time is distorted by outliers (queue delays, emergency holds). p75 gives a realistic, conservative bound for scheduling.

FALLBACK VALUES (WHEN PAIR NOT OBSERVED)

Same-floor tasks	5 min
Cross-floor tasks	8 min
p90 — all pairs	27.5 min

57.0% of O-D pairs cannot complete the full task within 15 min by travel time alone — confirming pickup response time as the correct KPI.

01

Travel Matrix

OR-Tools VRPTW routing cost input

02

Task History

RAG knowledge base · K=10 similar tasks

03

Labeled Tasks

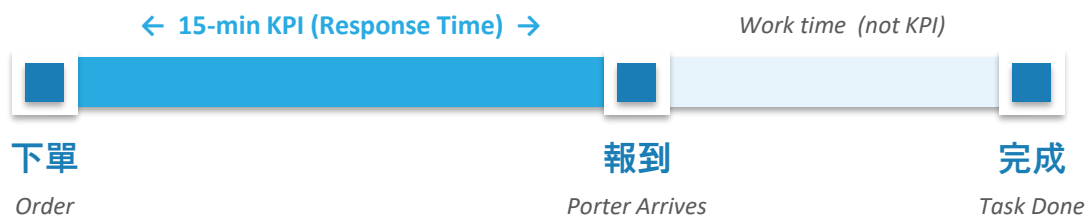
NLP parsing accuracy validation · n=100

What Does the "15-min KPI" Actually Mean?

KPI DEFINITION

KPI = Pickup Response Time

NOT total task completion time



WHY PICKUP RESPONSE TIME?

Dispatch algorithms control when a porter departs, not how long travel takes between locations. Total completion time conflates geography with dispatch quality.

57.0% of origin-destination pairs cannot physically complete the full task within 15 min by travel time alone.

Total duration is geography-constrained; pickup response is dispatch-controllable.

Algorithm cannot reduce travel distance; dispatch can only control WHO arrives WHEN.

TWO METRICS — ONE SYSTEM

75%

Total Duration Violation

historical measure — not our KPI

41.3%

Pickup Violation

our optimization target

WHAT DISPATCH CAN AND CANNOT CONTROL

Dispatch CAN Improve

- ✓ Pickup response time
- ✓ Porter selection & routing
- ✓ Queue management strategy

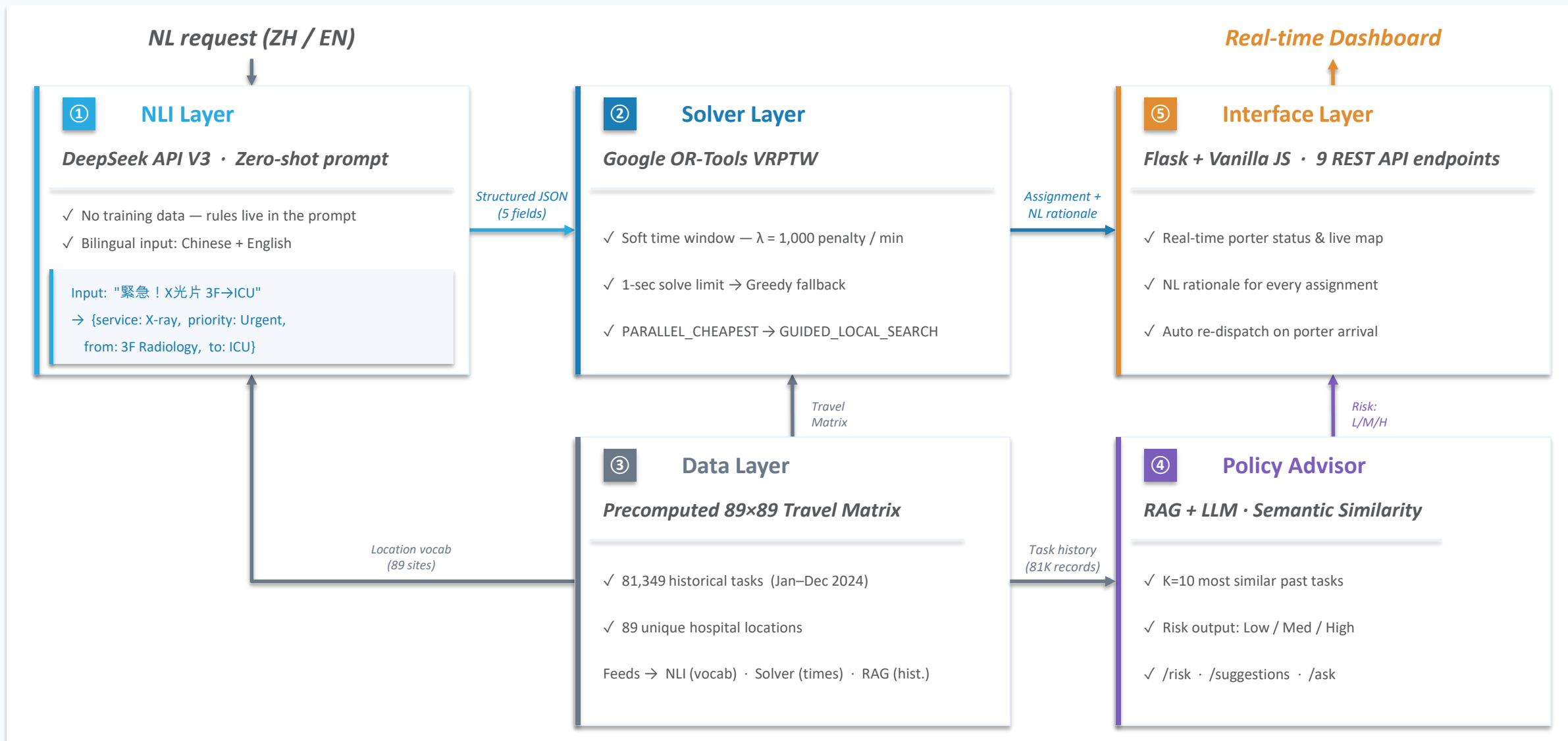
Geography Constrains

- ✗ Total task duration
- ✗ 57% routes: travel > 15 min total
- ✗ No algorithm reduces distance

Pickup response time is the only metric an intelligent dispatch algorithm can improve.

41.3% baseline is the reference for all results in this presentation.

System Architecture — Hybrid AI Dispatch



Component 1 — LLM Natural Language Interface

NL → JSON CONVERSION

EN

"Super urgent patient from A&D to SDU"



```
{"from": "A&D", "to": "SDU", "service": "admission", "priority": "Super-Urgent"}
```

中

"緊急！X光片從3樓放射科送至消毒房"



```
{"from": "3/F Radiology", "to": "Sanitizing Room", "service": "X-ray", "priority": "Urgent"}
```

ZERO-SHOT DESIGN

- ✓ No training data required — rules live in the prompt
- ✓ Handles unseen phrasings automatically
- ✓ Easy to update: change the prompt, not retrain

COST COMPARISON (PER 1M TOKENS)

GPT-4o

\$2.50*per 1M tokens*

DeepSeek v4 flash ✓

\$0.14*per 1M tokens*

~18× cheaper · Low call volume makes cost negligible at HHH scale

EXTENDED CAPABILITIES

- ✓ Chinese + English bilingual input
- ✓ Multi-task from a single request
- ✓ Intermediate stops / waypoints
- ✓ Scheduled future tasks
- ✓ Infection control flags (VRE, NEATS)
- ✓ LLM "Why this porter?" rationale output

Component 2 — OR-Tools VRPTW Optimizer

VEHICLE ROUTING PROBLEM — NODE GRAPH

m Porter Start Depots

Porter A · Porter B · Porter C · ...



n Pickup Nodes (task origins)

Pickup 1 · Pickup 2 · ... · Pickup n



n Delivery Nodes (task destinations)

Delivery 1 · Delivery 2 · ... · Delivery n



End Depot (dummy — all routes terminate here)

Objective Function

$$\min \left\{ \sum_{i,j} c_{ij} x_{ij} + \lambda \sum_k \max(0, \text{response}_k - 15 \text{ min}) \right\}$$

↑ travel cost

↑ KPI soft penalty ($\lambda = 1,000$ centi-min per min over target)

Hard: capacity = 1 · pickup before delivery enforced per porter

CORE CONSTRAINTS

①

Capacity = 1 task per porter at a time

②

Pickup before delivery — same porter enforced

③

Drop penalty = $2 \times 10^7 \gg$ any KPI penalty

④

Soft time window: $\lambda = 1,000$ per min over 15-min KPI

SEARCH STRATEGY

Step 1: PARALLEL_CHEAPEST_INSERTION

▼ initial solution

Step 2: GUIDED_LOCAL_SEARCH (1-sec time limit)

Fallback: Greedy (nearest available porter when solver times out)

ROLLING-HORIZON RE-OPTIMISATION

trigger: new task arrives

trigger: task completed

▼ Re-optimizes entire pending queue — ensures fresh dispatch at every event

Component 3 — RAG Policy Advisor

RAG RETRIEVAL PIPELINE

New task request → find similar past cases → LLM synthesises risk output

① **New Task Input**
service · origin · destination · time-of-day

② **Weighted Similarity Search**
81,349 cleaned records · 4-field scoring

③ **K=10 Retrieval**
top-10 most similar past tasks selected

④ **LLM Synthesis**
DeepSeek summarises patterns + assesses risk

⑤ **Risk Output + Recommendation**
Low / Medium / High · actionable policy suggestion

SIMILARITY WEIGHTS

+4 **Service Type**
Strongest KPI predictor

+2 **Destination**
Route-specific congestion

+2 **Origin**
Route-specific congestion

+1 **Time-of-day**
Shift load variation

3 ADVISOR MODES

/advisor/risk

KPI Risk Assessment

Scores incoming task vs K=10 similar records
→ outputs pickup violation rate + risk level

HIGH RISK · 3F X-ray, 2 porters busy
hist. pickup violation: 88.2% · Suggest: pre-assign

LOW

MEDIUM

HIGH

/advisor/suggestions

Strategic Staffing Advice

Aggregates patterns across active tasks
→ fleet-level service recommendations

3F X-ray: mean 177 min, 88.2% violation.
Consider dedicated porter or task batching.

/advisor/ask

Free-text Policy Q&A

Natural language queries over full history
→ 81,349-record knowledge base

Q: "Which service type needs more porters?"
A: 3F X-ray, 88.2% violation. Consider dedicated porter.

System in Action — End-to-End Demo

STEP 1 Natural Language Input

"Super urgent patient from A&D to SDU"

Chinese or English — both supported



STEP 2 LLM Parsing → Structured JSON

```
{"from": "A&D", "to": "SDU",  
  "service": "admission", "priority": "Super-Urgent"}
```



STEP 3 Optimizer Assigns + Rationale

→ Porter A · ETA: 4.2 min (example)

"Porter A is nearest free porter (4.2 min travel, example).
Infection flag: cleared — standard assignment."

FUNCTIONAL VALIDATION — 7 Scenarios Tested End-to-End (Table 6)

Multi-task

"Send a patient from 7H to A&D and a specimen from 5L to Lab"

→ 2 tasks structured → dispatched to 2 porters in single call

Infection Flag

"NEATS patient from 3C to A&D, VRE positive"

→ infection_control populated → red warning card surfaced to dispatcher

Chinese / English

"緊急 patient from 3FXRAY to 6H"

→ Chinese urgency token parsed → priority correctly classified as Urgent

Scheduled Task

"Send a patient from 7H to A&D at 14:30"

→ scheduled_at extracted → task queued, released at 14:30 with UI countdown

Multi-stop

"Send a patient from 7H to 3FXRAY, then continue to ICU"

→ stops=[3FXRAY] extracted → solver routes through intermediate stop to ICU

✓ Chinese & English NL input

✓ Multi-task from one request

✓ Intermediate stops / waypoints

✓ Infection control flags (VRE, NEATS)

✓ Scheduled future tasks

✓ Rolling-horizon re-optimisation

✓ LLM "Why this porter?" explanation

✓ RAG policy advisor

✓ Undo last dispatch

Video

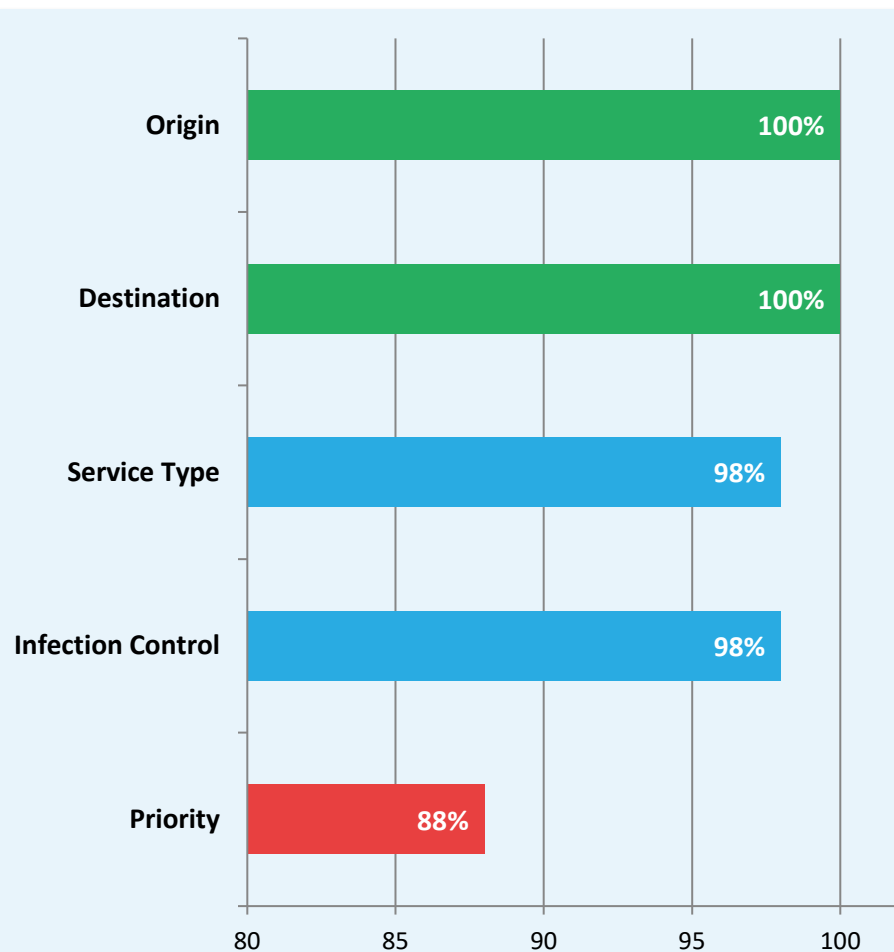


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How Accurate Is the LLM Parsing?



Priority errors: 即時 (Urgent) mis-classified as Normal — fixable with one prompt edit
X-axis starts at 80% to magnify differences

84%

all-field accuracy

5 out of 5 fields correct · n = 100 samples

Zero LLM parse failures

✓ Location: 100% Accuracy

Origin · 100% Destination · 100%

Correct hospital location parsed in every case.

Service Type · 98% Infection Flag · 98%

⚠ Priority: 88% — What went wrong?

12 errors

即時 (Urgent) mis-classified as Normal

Pattern

Systematic — not random noise

Root cause

Ambiguous Chinese term in zero-shot prompt

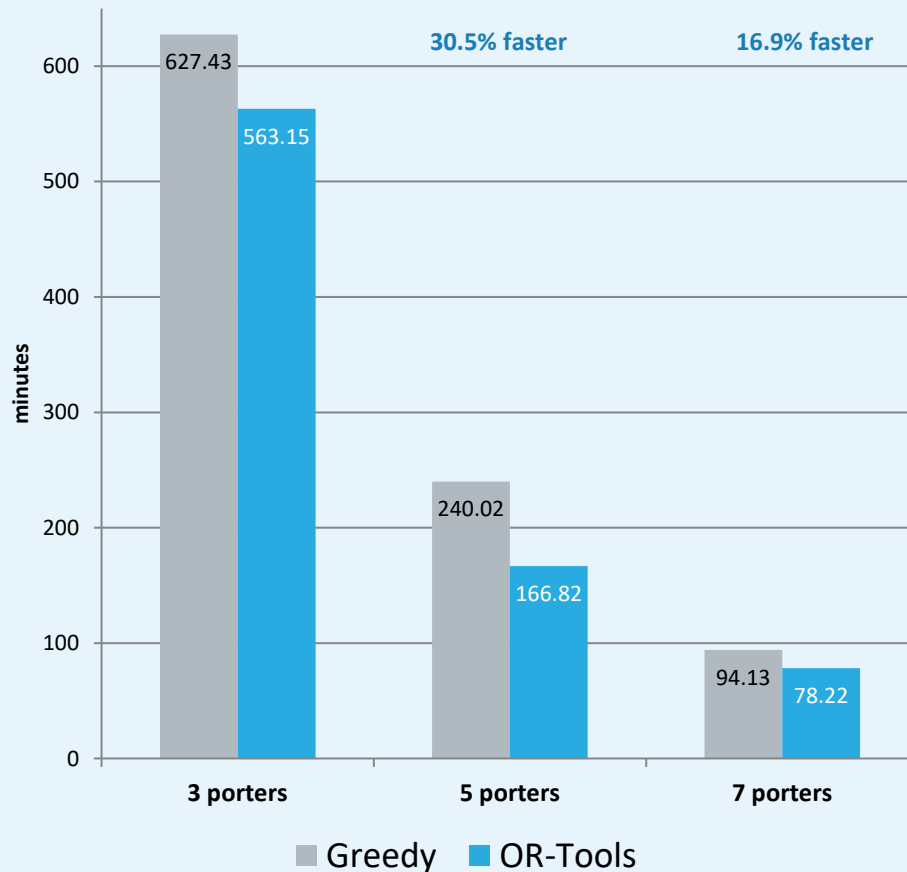
Resolution

1-line prompt clarification → resolvable

Does the Optimizer Actually Help?

Validated via discrete-event simulation:
synthetic (Poisson, 8-min inter-arrival) + historical replay

Mean Total Task Duration (min) — Synthetic Mode



Key Finding

OR-Tools helps most when the system is not saturated

Under Light Load (Synthetic, Poisson arrivals)

5 Porters

Greedy 240.0 min → OR-Tools 166.8 min

+30.5%

7 Porters

Greedy 94.1 min → OR-Tools 78.2 min

+16.9%

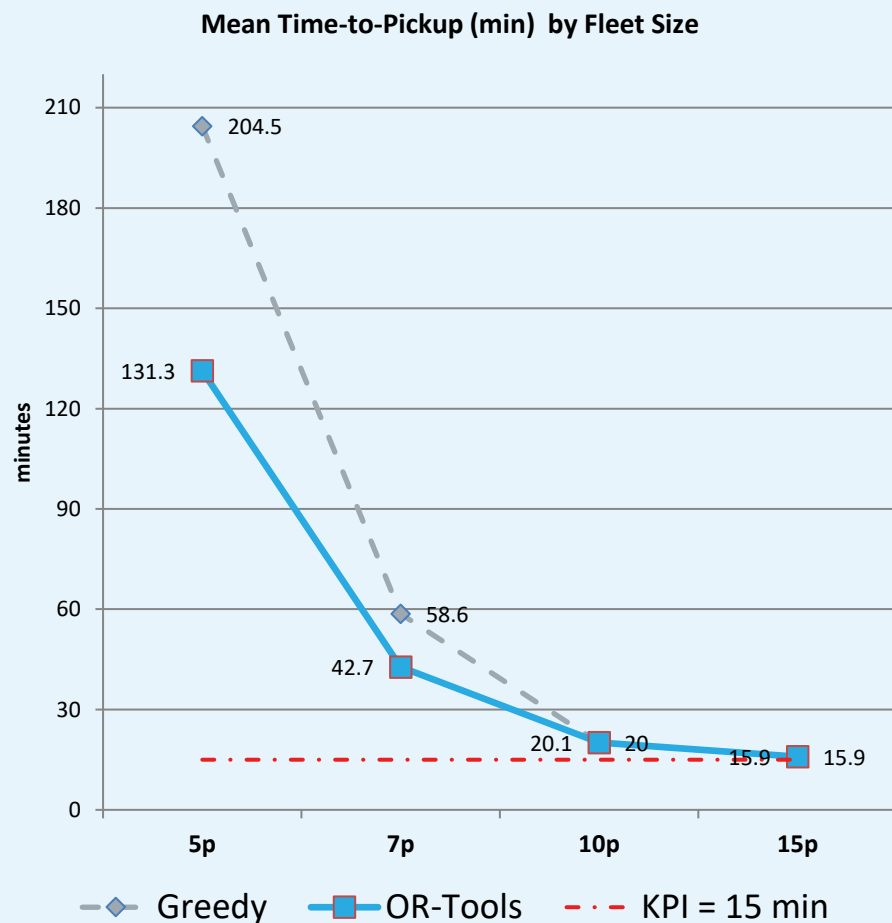
Under Heavy Load (Historical Replay)

Replay 5p: Greedy 158.6 min → OR-Tools 155.1 min (+2.2%)

Greedy already near-optimal under saturated conditions.

Marginal gain — system bottleneck is fleet size, not algorithm.

Fleet Sizing — How Close Can We Get to the KPI?



X-axis: evenly spaced (5, 7, 10, 15 porters) | At 15p both strategies converge: 15.9 min $\approx$ KPI

35.8%

faster pickup

5p OR-Tools vs Greedy

27.1%

faster pickup

7p OR-Tools vs Greedy

Fleet Sizing Recommendation

15 porters + OR-Tools

Mean Pickup: 15.9 min $\approx$ KPI Target (15 min)

15.9 min

mean pickup time

48%

violation rate

$\approx$ KPI

15-min target met

⚠ Management Takeaway

Algorithm alone cannot compensate for understaffing.

15 porters is a prerequisite — not a recommendation.

Summary of Contributions

①

Validated Natural Language Interface

- LLM-based parsing · DeepSeek API (zero-shot)
- 84% all-field accuracy · 100% location accuracy
- Bilingual (English + Chinese) · zero parse failures

②

OR-Tools VRPTW Optimizer

- 35.8% pickup time reduction (5p) · 27.1% (7p)
- 15 porters + OR-Tools → mean 15.9 min ≈ KPI
- Rolling-horizon re-optimisation at every event

③

Operational Intelligence Layer

- RAG policy advisor · 81,349 historical records
- KPI risk assessment (Low / Medium / High)
- Free-form Q&A over task history

DELIVERABLES

- ✓ Deployed web application (Flask + REST API)
- ✓ 89 × 89 travel time matrix (p75 percentile)
- ✓ Discrete-event simulator (Poisson + replay)
- ✓ NLP accuracy harness (n = 100 labelled samples)
- ✓ RAG policy advisor (K = 10 retrieval)
- ✓ Benchmark results (Greedy vs OR-Tools)

FUTURE WORK

- Real hospital deployment & live testing
- Annotated NLP dataset for fine-tuning
- Multi-hospital network expansion

Thank You

Q & A

OR-Tools Formulation Details

NODE GRAPH

- m porter start depots
- n pickup nodes (task origins)
- n delivery nodes (task destinations)
- 1 dummy end depot

OBJECTIVE FUNCTION

$$\min \left\{ \sum_{i,j} c_{ij} x_{ij} + \lambda \sum_k \max(0, \text{response}_k - 15 \text{ min}) \right\}$$

c_{ij} = travel time between nodes i and j (89×89 matrix)

λ = 1,000 centi-min per minute of KPI violation

W = 15 min (KPI soft time window)

CONSTRAINTS

Pickup before delivery (same porter)

Capacity: 1 task per porter at a time

Drop penalty: $2 \times 10^7 \gg$ any KPI penalty

Solve time limit: 1 second

SEARCH STRATEGY

① PARALLEL_CHEAPEST_INSERTION

First-solution construction

② GUIDED_LOCAL_SEARCH

Iterative improvement (1-sec limit)

Fallback

Greedy (nearest available porter)

Why Not Train a Custom NLP Model?

OPTION A — TRAINED CLASSIFIER

- ✗ Requires labeled training data (81,349 records to annotate)
- ✗ Retraining needed for any domain update
- ✓ Potentially higher ceiling accuracy

OPTION B — LLM ZERO-SHOT ✓ CHOSEN

- ✓ No annotation required
- ✓ Handles unseen phrasings automatically
- ✓ Easy to update — change the prompt, not retrain
- ✗ API cost per request
- ✗ Priority field: 88% accuracy (weakest field)

ZERO-SHOT ACCURACY (n = 100, zero parse failures)

84%all fields
correct**100%**location
accuracy**88%**priority
accuracy

Source: Table 1 — field-by-field accuracy on 100 historical test samples

VERDICT

For a hospital prototype without labelled data, the LLM approach is the right engineering choice.

84% all-field accuracy is sufficient for dispatch with human oversight.

A trained model is the natural next step if HHH commits to full deployment and provides annotated records.

Pickup-Time KPI Analysis — Table 5

KPI = 15 min pickup response time

Porters	Strategy	Mean Pickup Time (min)	KPI Violation Rate	Pickup Improvement	Note
5	Greedy	204.5	96.0%	— (baseline)	
	OR-Tools	131.3	83.0%	v 35.8% faster	
7	Greedy	58.6	94.0%	— (baseline)	
	OR-Tools	42.7	78.0%	v 27.1% faster	
10	Greedy	20.0	59.0%	— (baseline)	
	OR-Tools	20.1	60.0%	~0% (converge)	Supply adequate — Greedy near-optimal
15	Greedy	15.9	48.0%	— (baseline)	KPI target met
	OR-Tools	15.9	48.0%	0.0% (match)	RECOMMENDED: 15+ porters

Source: Table 5 — synthetic simulation (Poisson arrivals, 8-min mean inter-arrival, 100 tasks/run) | KPI = pickup response time (order placed -> porter at origin) | No Median / SD reported in current simulation output

Full Simulation Data — Raw Results

SYNTHETIC MODE (Poisson 8-min inter-arrival)

#p	Strategy	Mean Duration	Mean Pickup Wait	Violation %	Completed
3	Greedy	627.4	569.4	100.0%	100/100
	OR-Tools	563.2	508.1	100.0%	100/100
5	Greedy	240.0	183.0	100.0%	100/100
	OR-Tools	166.8	114.5	100.0%	100/100
7	Greedy	94.1	37.1	100.0%	100/100
	OR-Tools	78.2	23.4	100.0%	100/100

All times in minutes | Mean Pickup Wait = pickup response time (KPI metric)

HISTORICAL REPLAY MODE (real HHH timestamps)

#p	Strategy	Mean Duration	Mean Pickup Wait	Violation %	Completed
3	Greedy	293.5	271.9	99.0%	100/100
	OR-Tools	286.6	265.3	99.0%	96/100
5	Greedy	158.6	137.1	97.0%	100/100
	OR-Tools	155.1	133.6	96.8%	94/100
7	Greedy	104.4	83.0	96.0%	100/100
	OR-Tools	103.4	82.2	95.8%	95/100

All times in minutes | Mean Pickup Wait = pickup response time (KPI metric)

Simulation Design Details

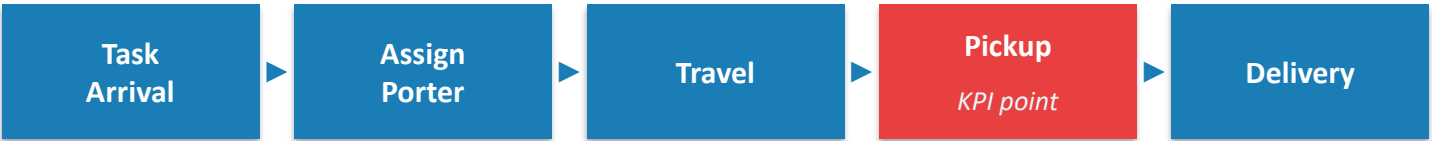
SYNTHETIC MODE

Arrivals	Poisson process · mean 8 min inter-arrival
Calibration	~10 tasks / hour / shift (from 2024 data)
Svc types	Distribution approximates historical data
Scale	100 tasks per run · 30 independent runs
Purpose	Controlled & repeatable — fleet-size scaling

HISTORICAL REPLAY MODE

Source	100 tasks from DATA2024.xlsx · chronological
Arrivals	Real inter-arrival times preserved
Load	Approximates single peak-shift workload
Runs	Single replay per date slice
Purpose	Actual HHH distribution — worst-case bound

EVENT FLOW & METRICS



← time-to-pickup (KPI: must be < 15 min) →

① **Total task duration**
order → delivery complete

② **Time-to-pickup**
order → porter at origin ← KPI

③ **KPI violation rate**
pickup time > 15 min